# PW-MFL: Promoting Semantic Segmentation in Resolution-Degraded Aerial Images via Pixel-Wise Mutual-Feed Learning

Jinze Yang, Youming Wu, Wei Dai, Wenhui Diao, Zining Zhu, *Member, IEEE*, Xian Sun, *Senior Member, IEEE*, and Kun Fu, *Member, IEEE*

***Abstract*— Due to variable imaging conditions, resolution degradation often occurs in aerial images, which in turn impairs the performance upper bound of semantic segmentation (SS). To solve this problem, super-resolution (SR) is placed before SS as a preprocessing subtask in most existing methods. The above two subtasks often form a unidirectional open-loop structure for relatively independent optimization, which constrains the ultimate segmentation performance improvement. To break down information barriers among them and form a more compact overall optimization, we propose an effective learning method named as pixel-wise mutual-feed learning (PW-MFL) for segmenting images with resolution degradation. The key is to build auxiliary bidirectional connections, which contribute to the mutual pixel-wise spatial and feature information guidance during training. The feed-forward connection is realized by the self-attention context correlation (SACC) module, which enhances the intraclass semantic features of pixel positions with poor SR performance by the fusion of that with superior performance. The feedback connection is achieved by the semantic weighted mapping (SWM) module, which aims to activate and adjust the detailed features of SR in incorrectly segmented pixel positions via the semantic feature information. In addition, the pixel-aware optimization (PAO) strategy is proposed to give more attention to optimizing specific pixel positions based on spatial information. Extensive experiments are conducted on three representative remote sensing segmentation benchmarks, ISPRS Vaihingen, ISPRS Potsdam, and iSAID datasets. The state-of-the-art (SOTA) segmentation level in resolution-degraded aerial images is achieved through the proposed learning method.**

***Index Terms*— Pixel-aware optimization (PAO), pixel-wise mutual-feed learning (PW-MFL), self-attention context correlation (SACC), semantic weighted mapping (SWM), spatial and feature information.**

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Jinze Yang, Wenhui Diao, Zining Zhu, Xian Sun, and Kun Fu are with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100190, China, also with the Key Laboratory of Network Information System Technology (NIST), Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100190, China, and also with the School of Electronic, Electrical and Communication Engineering, University of Chinese Academy of Sciences, Beijing 100190, China (e-mail: yangjinze20@mails.ucas.ac.cn; diaowh@aircas.ac.cn; zhuzining20@mails.ucas.ac.cn; sunxian@aircas.ac.cn; kunfuiecas@gmail.com).

Youming Wu and Wei Dai are with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100190, China, and also with the Key Laboratory of Network Information System Technology (NIST), Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100190, China (e-mail: youming wu@yeah.net; daiwei@aircas.ac.cn).

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## NOMENCLATURE

| Abbreviations | The Meanings of the Abbreviations |
|---|---|
| SS | Semantic segmentation. |
| SR | Super-resolution. |
| CIs | Clear images. |
| RIs | Restored images. |
| RDIs | Resolution-degraded images. |
| SSGM | SS guidance mask. |
| SRGM | SR guidance mask. |
| PVD | Pixel value difference. |
| SACC | Self-attention context correlation. |
| SWM | Semantic weighted mapping. |
| PAO | Pixel-aware optimization. |

## I. INTRODUCTION

IN THE remote sensing field, SS is vital for image interpretation [1], [2], [3], [4], which is widely implemented in road extraction, urban change detection, scene understanding, and other multifarious applications. In existing methods [5], [6], [7], [8], CIs with high resolution are indispensable to obtain satisfactory performance [9], [10]. However, CIs are often consuming to capture [11], [12], [13]. The SS performance in the relevant cases still needs to be improved due to the decreasing image resolution. As shown in Fig. 1, the edge and small object information are scarcely available in RDIs, and the precise segmentation is intractable to achieve.

To resolve the issue, the SR technology has been widely performed to enhance RDIs for SS. It aims to restore detailed texture information in RDIs. Guo et al. [14] conducted the training of the SR and SS model independently, which limits the final segmentation performance. Xu et al. [15] proposed an end-to-end mode to train the two tasks just with the segmentation supervision. Wang et al. [9] further designed a parallel framework to train the two tasks with a multitask loss optimization. These two methods both exploit an overall loss function to jointly optimize the two tasks for better segmentation results. However, just utilizing a joint optimization based on an open-loop structure is a loosely coupled approach to connect the two tasks, and the information of each task is still in a self-contained state. The intertask information barrier fails to be broken. Therefore, the results obtained through the overall optimization of the two tasks are still limited, which in

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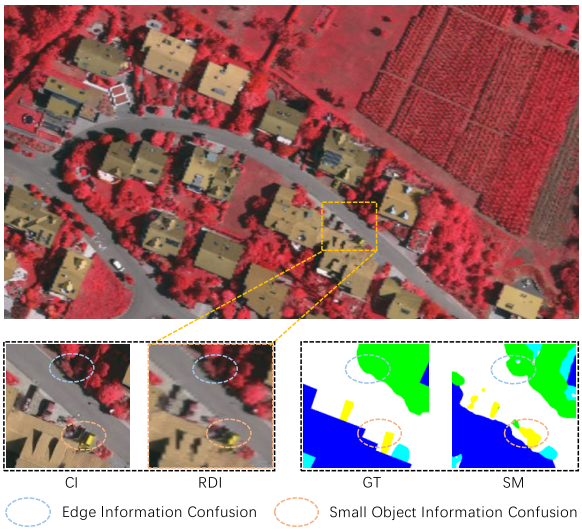



Fig. 1. Illustrations of segmentation challenges brought by RDI. 1) Edge information confusion (indigo blue circle). 2) Small object information confusion (pale red circle). CI is the clear image; RDI denotes the resolution-degraded image; GT means the ground truth; and SM represents the segmentation mask.

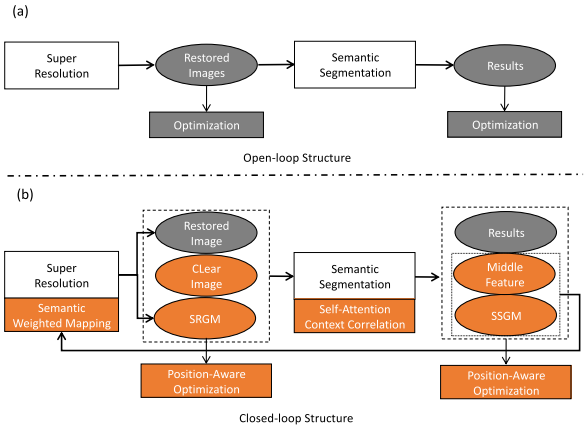



Fig. 2. Illustrations of the whole method for the training stage. (a) **Open-Loop Structure:** In the previous method, only the RIs are inputted into SS. (b) **Closed-Loop Structure:** In the proposed method, two masks are generated and inputted into the two networks. The extra feature operations are conducted with spatial guidance. In the inference stage, the orange parts are not utilized.

turn constrains the ceiling of segmentation accuracy in RDIs. In short, the optimization of each task should adapt to that of another task, and the in-depth information interaction should be exploited.

According to the above analysis, SR and SS are optimized jointly just built on a simple open-loop structure. The in-depth information interaction between the two tasks is ignored. Specifically, the missing in-depth information interactions that limit the optimization effect are described in the following. For one thing, there exist spatially variant PVD cases between RIs and CIs. Without being aware of this phenomenon, it poses a huge challenge for SS models to extract a uniform feature representation for each class. For another, due to missing the precise spatial information guidance of wrong segmentation pixel positions, it is difficult to obtain the optimal RI required by SS precisely in SR. Furthermore, balancing the weight of two different-level tasks is challenging in the multitask loss design [16], which directly affects the segmentation performance.

Due to the existing problems, the degree of coupling between the two tasks is weak, and the segmentation performance is limited. The specific pixel-wise spatial information in the results of each task should be exploited to boost the segmentation. Therefore, we design a new learning method to simultaneously optimize the two tasks. It is named as pixel-wise mutual-feed learning (PW-MFL). Built on the conventional open-loop structure, it is expected to add bidirectional connections to form a closed-loop structure, where the pixel-wise spatial and feature information can be fed mutually as auxiliary guidance between the two tasks in the training stage. Then, based on the guidance, precise operations are performed at specific pixel positions to enhance the features of the two tasks. Meanwhile, in the inference stage, only the original open-loop structure is utilized to obtain better segmentation performance without increasing the model complexity.

Different from the existing region-wise guidance proposed for the object detection task, the pixel-wise guidance is exploited specifically for the SS task in the article. The critical points of the proposed method are as follows. The mutual pixel-wise spatial information guidance is established between the two tasks. This involves the generation of an SRGM based on the PVD cases between RIs and CIs. In addition, an SSGM is formulated by analyzing the pixel-wise result between the segmentation mask and the GT. Based on the guidance, an SACC module is introduced in SS to enhance features of pixel positions with higher PVDs. Meanwhile, a SWM module is designed in SR to incorporate the semantic information into the restoration of wrong segmentation pixel positions. Then, a PAO strategy is introduced to refine the optimization process by leveraging the spatial information guidance. This strategy allocates specific attention to the mentioned pixel positions. Moreover, the two tasks are optimized iteratively to mitigate the negative effect caused by the improper weight distribution. The core implementation is shown in Fig. 2.

To sum up, the innovative contributions are summarized in the following.

1) To enhance the interaction and break down information barriers between the two tasks, we propose a new learning method named PW-MFL for training SR and SS simultaneously.
2) Through analyzing the discrepancy between the prediction results and the GT, two masks, SRGM and SSGM, are generated and utilized to provide pixel-wise spatial information as guidance.
3) A SACC module and a SWM module are proposed to perform pixel-wise feature enhancement in SS and SR with pixel-wise guidance, respectively.
4) To refine the optimization direction, a PAO strategy is designed to allocate additional attention to particular pixel positions of spatial information.

Extensive experiments are conducted on three datasets, ISPRS Vaihingen, ISPRS Postdam, and iSAID. The satisfactory results show the effectiveness and generalization of the

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proposed method. The remainder of this article is structured as follows. A concise overview of the related work is provided in Section II. Section III offers the implementation details of our proposed PW-MFL method, including the feed-forward and feedback connections. Experiments were conducted on Vaihingen, Potsdam, and iSAID datasets and the corresponding results are demonstrated in Section IV. The article concludes with a summary in Section V.

## II. Related Works

In the existing methods, SR and SS have been extensively studied. Next, we describe the relevant works from the following three aspects.

### A. Semantic Segmentation

There are many well-behaved segmentation models [5], [17], [18], [19], [20], [21], [22] for SS of various tasks. Deeplabv3+ [19] is proposed by exploiting the atrous separable convolution to expand the receptive fields. PSPNet [7] is proposed to extend the receptive field by exploiting atrous separable convolutions. CCNet [6] is proposed by designing a cross-attention module to extend the receptive field. EANet [23] is proposed by developing a novel attention mechanism called external-attention which is based on two external memories to explore potential correlations between different samples. Niu et al. [24] utilized a graph convolution module to mine the context information based on a disentangled learning strategy. HSDN [25] decouples the high-order semantic feature to enhance the intraclass uniform.

To speed up the inference stage, there exist methods for designing compact models [26], [27], [28] to speed up the inference stage. ENet consists of different compact bottlenecks in different parts. ERFNet [29] is proposed by using the residual structure based on ENet [26], the decomposed convolution, and the dilated convolution. ESPNet [30] is developed through a module with the pyramid structure and the dilated convolution. LiANet [31] contains two enhanced attention modules to further optimize the generated attention map.

### B. Super-Resolution

The powerful nonlinear modeling capability of CNN provides great possibilities for SR [32], [33], [34]. EDSR [33] utilizes enhanced residual structure for SR. DDBPN [35] contains an error feedback mechanism through upsampling and downsampling layers. RCAN [36] is developed by designing a residual channel attention (RCA) module to capture the channel-range interaction. With continuous innovation in the image generation field, generative adversarial network (GAN) [37], [38], [39], [40] shows a more robust ability for SR. GAN consists of a generator that generates RIs with the input of RDIs, and a discriminator that distinguishes whether the input images are authentic or forged. SRGAN [38] is developed by utilizing a perceptual loss function to pay close attention to the feature-level restoration. There also exist many methods [8], [41], [42], [43], [44], [45] for SR in many tasks with data of different types.

### C. SR for SS

Due to the negative effect brought by resolution degradation for SS, there exists research utilizing SR to improve the input quality of SS. Guo et al. [14] utilized SR to resolve the resolution reduction before performing segmentation on new data. Cui et al. [46] proposed two-stage training of the two tasks for better green tide extraction by utilizing restored more high-resolution information. Xu et al. [15] proposed an end-to-end training mode and just utilized the segmentation supervision. Aakerberg et al. [47] utilized the segmentation loss to guide SR based on a domain adaptation. Wang et al. [9] incorporated the restored features of the SR branch into the SS branch, which can utilize more detailed features for better segmentation. Zhang et al. [10] developed a collaborative network using multiscale feature fusion for SR, and utilized SSIM as the feature supplementation module. Zhang and Zhang [48] utilized SR to enhance the features in the last upsampling layer of SS. Salgueiro et al. [49] incorporated multistage features of SR into SS to further improve the segmentation performance.

It can be found that existing studies for jointly training SR and SS based on open-loop systems just rely on the overall loss function. This method can achieve a certain degree of information interaction, but can not completely break the information barriers between the two tasks. The two tasks are in a loosely coupled state, and the two tasks cannot sense each other's current training results and adjust the training stage accordingly. In this article, to solve the problem, additional feed-forward and feedback connections are designed to further improve the information interaction between the two tasks and make the two tasks tightly coupled. The pixel-wise spatial and feature information is exploited as guidance to enhance the training stage of each other.

## III. Methodology

The previous methods for performing segmentation in RDIs utilize the open-loop structure. The in-depth correlation between the two tasks is not fully exploited. The simple feed-forward connection just feeds RIs to the subsequent segmentation, which is too weak to obtain better final segmentation performance. The optimization of two subtasks should adapt to each other. Therefore, considering the results of the two subtasks are relevant, the optimization of each subtask should be tailored to the results of the other subtask. The auxiliary pixel-wise spatial information that needs to be exploited in the results is fed to each other, which makes the two subtasks tightly coupled by designing a closed-loop structure. Then, the pixel-wise adjustment operation is performed to optimize the features in both subtasks. Therefore, the segmentation results are improved through the proposed method. For clarity, we have summarized the main abbreviations as demonstrated in Nomenclature.

### A. Approach Overview

In this section, the details of the proposed PW-MFL method are described. It consists of two subtasks, SR and SS. There are two structures for the training and inference stages,

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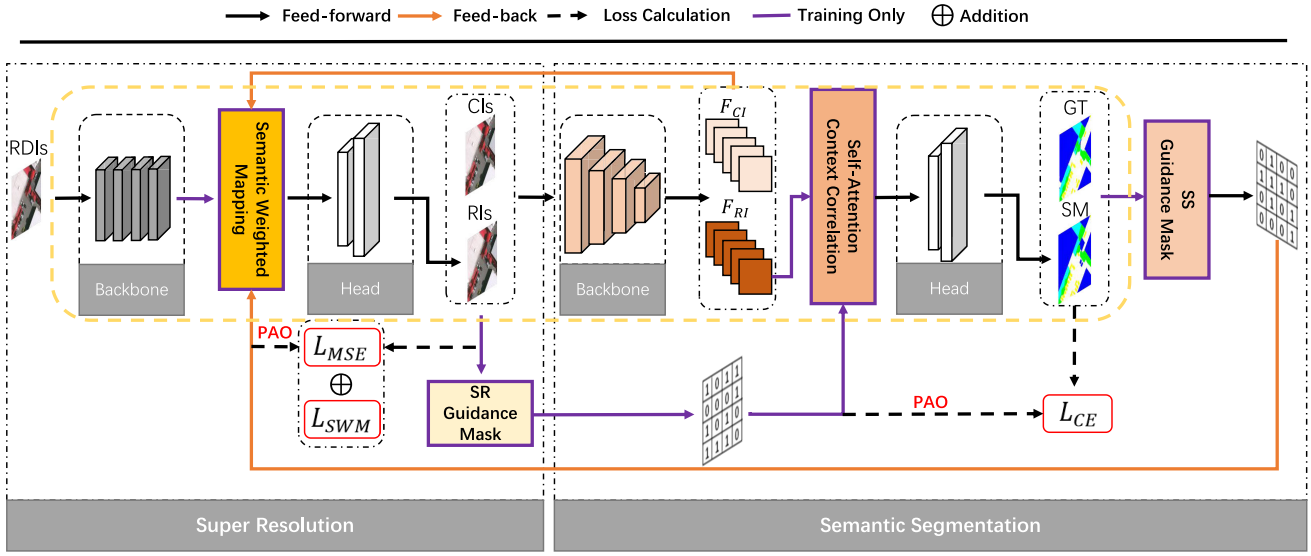



Fig. 3. Whole pipeline of simultaneously training SR and SS in this article. It contains the feed-forward connection as the black line and the feedback connection as the orange line. PAO means the pixel-aware optimization; RDIs mean resolution-degraded images; CIs mean clear images; RIs mean restored images; SM means the segmentation mask; GT means the ground truth; $F_{\mathrm{CI}}$ is on behalf of the output feature of CIs; $F_{\mathrm{RI}}$ represents the output feature of RIs; and $L_{\mathrm{SWM}}$ is loss proposed in the SWM module. Except for the purple components, other components of feed-forward connections in the yellow dotted box are utilized in the inference stage. Meanwhile, CIs are also not inputted into the SS model in the inference stage.

**Algorithm 1** Pixel-Wise Mutual-Feed Learning

**Require:** The SR model $f(\cdot|\theta_{SR})$, the SS network $f(\cdot|\theta_{SS})$, the SS dataset containing RDIs.

**Ensure:** RIs, Segmentation masks.

1: Input RDIs into $f(\cdot|\theta_{SR})$ and obtain RIs;
2: **repeat**
3: Generate SRGM as formula (2);
4: **Step 1: feed-forward Connection:**
5: Input SRGM, RIs, and CIs to $f(\cdot|\theta_{SS})$;
6: $F_{RI}$ and $F_{CI}$ are extracted;
7: Process $F_{RI}$ in SACC Module and obtain enhanced features $\hat{F}_{RI}$;
8: Perform PAO for the optimization of SS as formula (11).
9: Generate SSGM as formula (12);
10: **Step 2: feed-back Connection:**
11: Input RDIs, SSGM and $F_{CI}$ to $f(\cdot|\theta_{SR})$;
12: Process $F_R$ in SWM Module and obtain enhanced features $\hat{F}_R$;
13: Perform PAO for the optimization of SR as formula (22).
14: **until** convergence

respectively. In the inference stage, only the basic open-loop structure is exploited, and the parameters of the structure are obtained from the training stage. The performance is further improved without increasing model complexity. The key point of PW-MFL is to enhance the training stage. Built on PW-MFL, auxiliary bidirectional connections are added to the basic open-loop structure. The pixel-wise spatial information extracted from the results of one subtask is utilized to guide the optimization of another subtask. The optimization of the two subtasks is tightly coupled based on a closed-loop structure design which further improves the final segmentation performance. The additional feed-forward connection is designed to transfer the spatial information of different pixel value (PVD) difference cases from SR to SS, which makes SS aware of the pixel-wise restoration situation to extract more uniform feature representations. The additional feedback connections are developed to feed the spatial and feature information related to the pixel-wise segmentation cases from SS to SR, with which the semantic-aware restoration is performed to obtain RIs required by SS precisely. In addition, the existing models for SR and SS are directly chosen to act as the baseline model to prove the effectiveness and generalization of the proposed method. The whole algorithm is summarized in Algorithm 1. $F_{\mathrm{CI}}$ and $F_{\mathrm{RI}}$ mean the extracted features by the backbone from CIs and RIs, respectively. $F_R$ means the restored features in SR.

### B. Closed-Loop Structure

As shown in Fig. 3, an additional feed-forward connection is built to feed the generated SRGM from SR to SS in the training stage. The spatial information of pixels with different PVD cases in SRGM is utilized to enhance the extracted features by designing an SACC module in SS. Similarly, two auxiliary feedback connections are designed to feed the generated SRGM and the semantic features of CIs from SS to SR. Built on these, the semantic awareness is implemented in specific pixel positions by designing a SWM module in SR. Then, a closed-loop structure is formed by proposing the above pixel-wise operations. Finally, a PAO strategy is performed with pixel-wise spatial information guidance, which allows auxiliary attention to specific pixel positions in the optimization of both tasks.

*1) Feed-Forward Connection:* To restore the texture details in RDIs, SR is utilized to enhance RDIs before segmentation.

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In existing methods, the features are extracted in RIs under the supervision of the GT. However, there are always spatially various PVDs between RIs and CIs. Pixels with higher PVDs may lack optical information about the object and are unable to provide sufficient features. Meanwhile, the optimization of the segmentation in these pixels becomes more difficult.

Therefore, the spatial information of pixels with higher PVDs can not be ignored to guide the optimization of SS. To make SS aware of the pixel-wise restoration cases, the SRGM is generated to provide pixel-wise spatial information by analyzing the disparity between RIs and CIs. Built on SRGM, an SACC module is designed to make up for the feature extraction in pixels with higher PVDs with class relationships. Meanwhile, the optimization of SS can give these pixels more attention via a PAO loss. Through the subsequent experiments, the effectiveness of the above components is evaluated and demonstrated.

*a) SR guidance mask:* In existing methods, RIs are only utilized to calculate the similarity of RIs and CIs for optimization. However, the extracted features contain different levels of information due to the different cases of restoration in each pixel. Specifically, there are spatially variant PVDs between RIs and CIs. An information gap is caused by PVDs in optical RS images. The suitable features are difficult to extract in pixels with higher PVDs for SS, which may lead to wrong segmentation results. Therefore, it is essential to use the spatial information of pixels with higher PVDs to guide the optimization of SS. Then, the SRGM is generated to provide the spatial information as follows.

The RIs are first generated by the SR model $f(\cdot|\theta_{\mathrm{SR}})$ with the RDIs input, and a matrix $M_{\mathrm{PVD}}$ is obtained by calculating PVDs between RIs and CIs as (1). $\rho_{\mathrm{SR}}$ is the parameters of the SR model

$$\begin{aligned} \mathrm{RIs} &= f(\mathrm{RDIs}|\theta_{\mathrm{SR}}) \\ M_{\mathrm{PVD}}(i,j) &= \frac{1}{C_I}\sum_{c=0}^{C_I} |\mathrm{RIs}(c,i,j) - \mathrm{CIs}(c,i,j)|. \end{aligned} \tag{1}$$

The $(i, j)$ represents the pixel position in images. $|\cdot|$ denotes the absolute value calculation function. $M_{\mathrm{PVD}}$ is adjusted from a 4-D tensor to a 3-D tensor first, specifically from $B \times C_I \times W_I \times H_I$ to $B \times C_I \times (W_I \times H_I)$. $B$, $C_I$, $W_I$, and $H_I$ denote the batchsize number, channel number, weight, and height of the input images, respectively. $(\cdot \times \cdot)$ denotes that the dimensions in parentheses are combined into one dimension.

To reflect the relative restoration case of each pixel, the values of $M_{\mathrm{PVD}}$ in the last dimension are mapped to the [0,1] interval by using a softmax function. Considering the values of $M_{\mathrm{PVD}}$ are various, the case of the restoration is complicated. The available spatial information is too complex to be utilized simply. To simplify the issue, a threshold is set as the mean value $m$ of $M_{\mathrm{PVD}}$. In the last dimension, if the $M_{\mathrm{PVD}}(i)$ is less than the threshold, $p_i$ is regarded as the lower middle PVD pixel, and the value is set as 1; otherwise, $p_i$ is regarded as the upper middle PVD pixel, and the value is set as 0. Then, the pixels in RIs are divided into two categories. The SRGM is specifically obtained as follows:

$$\mathrm{SRGM}(i) = \begin{cases} 1, & M_{\mathrm{PVD}}(i) < m \\ 0, & M_{\mathrm{PVD}}(i) \geq m. \end{cases} \tag{2}$$

In the subsequent experiments, we analyze the choice of thresholds and verify that the mean is a better choice to obtain higher segmentation performance.

Finally, the size of SRGM is adjusted to $B \times 1 \times (W_I \times H_I)$. Therefore, pixels with different PVDs are distinguished through SRGM, and pixel-wise restoration cases are provided.

*b) Self-attention context correlation:* The most intuitive reason for the segmentation performance gap between RIs and CIs is the existing PVDs between RIs and CIs. The SS model is unable to adapt well to this situation to learn a better feature representation for each class. In existing methods, context information aggregation methods [24], [50], [51], [52] play a significant role in achieving satisfactory results by exploiting the intraclass correlation. Therefore, we utilize context information to enhance the extracted features based on the pixel-wise spatial information in SRGM. In particular, a SACC module is designed to achieve this purpose. It is considered that the upper middle PVD pixels miss more optical information about the class and may not provide insufficient features to get precise segmentation. On the contrary, the lower middle PVD pixels behave better in this aspect. Therefore, the analysis of the intraclass relationship between upper middle and lower middle PVD pixels is conducted to enhance the extracted features. Specifically, according to the same-class relationship, the relatively practical features of the lower middle PVD pixels are transferred to adjust the features of the upper middle PVD pixels. The implementation pipeline is shown in Fig. 4, and the implementation details are introduced as follows.

First of all, SRGM is resized to the scale of the features $F_{\mathrm{RI}}$ to perform the subsequent operations. The features $F_{\mathrm{RI}}$ with size $B \times C \times W \times H$ are inputted into SACC. $C$, $W$, and $H$ represent the class number, the width, and the height of the features, respectively. Then, the features of lower middle and upper middle PVD pixels are separated from $F_{\mathrm{RI}}$ with SRGM as (3). For clarity, the features of lower middle and upper middle PVD pixels are denoted as $v_2$ and $v_1$, respectively,

$$\begin{aligned} \mathrm{SRGM}' &= 1 - \mathrm{SRGM} \\ v_1 &= \mathrm{SRGM}' \cdot F_{\mathrm{RI}} \\ v_2 &= \mathrm{SRGM} \cdot F_{\mathrm{RI}}. \end{aligned} \tag{3}$$

It is expected that the intraclass valuable information of $v_2$ is delivered to augment $v_1$, and the better segmentation of each class is achieved. Therefore, built on the self-attention mechanism and context information, the same-class relationship is utilized to enhance $v_1$. In particular, the GT for segmentation is used to introduce the class information. The SRGM and GT are both resized into $B \times 1 \times (W \times H)$ for the calculation as (4), respectively. Therefore, the GT of upper middle and lower middle PVD pixels are obtained and regarded

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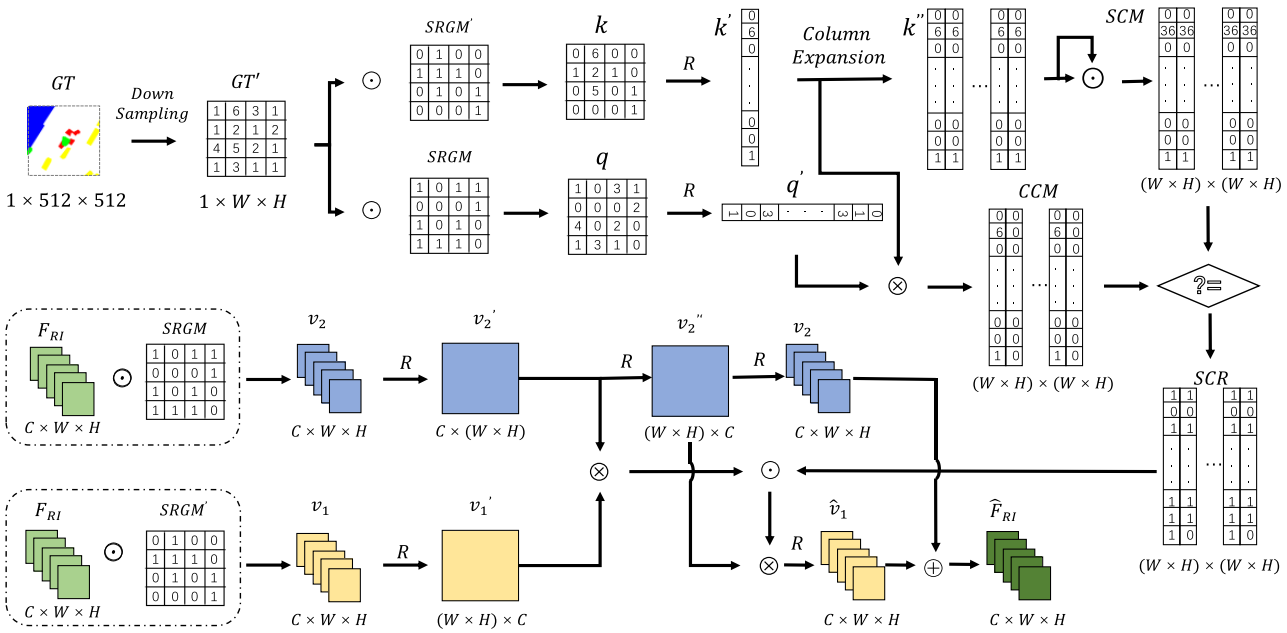



Fig. 4. Concrete implementation of SACC. The "R" operation is the abbreviation of the "Reshape." The "·," "×," and "+" symbols in a circle denote the element-wise multiply, the matrix multiply, and element-wise addition, respectively.

as $k$ and $q$, respectively,

$$k = \text{SRGM}' \cdot \text{GT}$$
$$q = \text{SRGM} \cdot \text{GT}. \tag{4}$$

Then two matrices are generated to obtain the intraclass relationship between pixel positions with upper middle and lower middle PVDs by utilizing $k$ and $q$. The implementation details are in the following.

$k$ and $q$ are resized to $B \times (W \times H) \times 1$ and $B \times 1 \times (W \times H)$ and denoted as $k'$ and $q'$, respectively. For each upper middle PVD pixel, pixels of the same class need to be found in low-middle PVD pixels. Then, a cross correlation matrix (CCM) is designed to establish the class correlation between $k'$ and $q'$. To demonstrate the case that all pixel pairs are of the same class between $k'$ and $q'$ as the reference, there also comes the same-class matrix (SCM). The two matrices are both with size $B \times (W \times H) \times (W \times H)$. Meanwhile, $k'$ is repeated $W \times H$ times in the third dimension to obtain $k''$ to calculate SCM. The specific calculation is as follows:

$$\text{CCM} = k' \times q'$$
$$\text{SCM} = k'' \cdot k''. \tag{5}$$

Therefore, the values of CCM represent the product results of the class label between each pixel of $k$ to all pixels of $q$. the values of SCM represent the product results in the case that all pixels of $q$ are of the same class as each pixel of $k$. Then, through the comparison between CCM and SCM, the pixel pairs of the same class between $k$ and $q$ are reserved in the same class relationship SCR matrix as follows:

$$\text{SCR}(i, j) = \begin{cases} 1, & \text{CCM}(i, j) = \text{SCM}(i, j) \\ 0, & \text{CCM}(i, j) \neq \text{SCM}(i, j). \end{cases} \tag{6}$$

If the value in CCM is equal to that in SCM, the value in SCR is set as 1; Otherwise, it is set as 0. So the matrix SCR represents the position of pixel pairs of the same class between $k$ and $q$: if the $i$th position in $k'$ is with the same class as the $j$th position in $q'$, SCR$(i, j)$ is 1; otherwise, SCR$(i, j)$ is 0. Therefore, the same-class relationship SCR matrix between upper middle and lower middle PVD pixels is obtained. Based on the position information in SCR, the features of upper middle PVD pixels are enhanced with lower middle PVD pixels for each class. In particular, the dimension of $v_1$ and $v_2$ are first adjusted into $B \times (W \times H) \times C$ and $B \times C \times (W \times H)$, and denoted as $v_1'$ and $v_2'$, respectively. We first calculate the feature correlation $C$ between $v_1'$ and $v_2'$ and find the related degree of the features between upper middle and lower middle PVD pixels. Then, $C'$ is obtained to reinforce the position of pixel pairs of the same class in $C$ based on SCR. Finally, $v_2'$ is adjusted into $B \times (W \times H) \times C$ and denoted as $v_2''$. Then, $C'$ is regarded as the weight to calculate the weighted sum of $v_2''$ adaptively for $\hat{v}_1$. Therefore, the features in lower middle PVD pixels are delivered to that of upper middle PVD pixels. The concrete calculation is illustrated in the following formula:

$$C = v_1' v_2'$$
$$C' = C \cdot \text{SCR}$$
$$\hat{v}_1 = C' v_2''. \tag{7}$$

Then, $\hat{v}_1$ and $v_2$ are resized to $B \times C \times W \times H$. The final enhanced features $\hat{F}_{\text{RI}}$ is updated as follows:

$$\hat{F}_{\text{RI}} = v_2 + \hat{v}_1. \tag{8}$$

Therefore, $\hat{F}_{\text{RI}}$ is utilized for the final prediction. By the proposed module, the features of upper middle PVD pixels are more related to that of the lower middle PVD pixels. The feature representation of each class is more uniform

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and effective for prediction. The attention is distributed more evenly in pixels of each class, whether in lower middle PVD or upper middle PVD pixels.

*c) PAO for SS:* The optimization of SS utilizes a cross-entropy (CE) loss with the extracted features $\hat{F}_{\mathrm{RI}}$ under the supervision of the GT. However, CE is operated on all pixels equally in the original methods. The wrong segmentation results are more likely to occur in upper middle PVD pixels. Therefore, more attention should be paid to upper middle PVD pixels in the optimization. To solve the problem, a new SS GT is generated based on the SRGM. Specifically, the lower middle PVD pixels are labeled as 255 in $\hat{\mathrm{GT}}$ based on GT. Pixels labeled as 255 are not taken into the calculation of CE loss, which ignores the optimization of these pixels. The original segmentation loss function is as follows:

$$\begin{aligned} S &= B \times C_s \times W \times H \\ \mathcal{L}_{\mathrm{Seg}} &= -\frac{1}{S}\sum_{n}^{B}\sum_{k}^{C_s}\sum_{i}^{W}\sum_{j}^{H}\mathrm{GT}(n,k,i,j)\log(\hat{F}_{\mathrm{RI}}(n,k,i,j)). \end{aligned} \tag{9}$$

$S$ means the total number of pixel positions where the segmentation loss is calculated. $C_s$ denotes the class number. $\hat{F}_{\mathrm{RI}}$ denotes the features utilized for the final prediction. $H$ and $W$ represent the height and width of the image input, respectively.

Based on the spatial information guidance, there comes the PAO loss as follows:

$$\begin{aligned} &\mathcal{L}_{\mathrm{PAO}} \\ &\quad = -\frac{1}{S}\sum_{n}^{B}\sum_{k}^{C_s}\sum_{i}^{W}\sum_{j}^{H}\hat{\mathrm{GT}}(n,k,i,j)\log(\hat{F}_{\mathrm{RI}}(n,k,i,j)). \end{aligned} \tag{10}$$

The value of $k$ is in $0, 1, \ldots, 5$ and is not equal to 255 for the mentioned purpose.

Built on the above calculation, we obtain the overall loss function to optimize SS as follows:

$$\mathcal{L}_{\mathrm{SS}} = \alpha * \mathcal{L}_{\mathrm{Seg}} + \beta * \mathcal{L}_{\mathrm{PAO}} \tag{11}$$

where $\alpha$ and $\beta$ mean the weight of traditional segmentation loss and that of PAO module loss in the overall SS loss. After utilizing the proposed optimization loss, the upper middle PVD pixels are assigned higher weights in the optimization process and given more attention to obtaining more precise optimization for better segmentation.

*2) Feedback Connection:* To cope with the decline of segmentation performance in the case of RDI input, we propose a new method. The final segmentation accuracy not only is dependent on the capability of the SS subtask but also limited by the restoration situation of the SR subtask. Similarly, the spatial information of wrong segmentation pixels not only is used to optimize SS itself but also to assist restoration in SR. Existing methods ignored the spatial information transfer which brings improvement to the final segmentation performance. To achieve this purpose, an SRGM is generated to provide the spatial information of the wrong segmentation results. Built on SSGM, the restored features in SR are enhanced with semantic features in CIs. The backward feedback connections are designed to guide the optimization of SR.

*a) SS guidance mask:* The SSGM is valuable in providing the spatial information of wrong segmentation pixels, which gives the wrong segmentation pixels more attention in SR for precise restoration. In particular, the discrepancies between the segmentation mask and GT are analyzed to obtain the wrong segmentation pixels. Then, the SSGM is generated to provide the spatial information. The operations for the generation of SSGM are the following.

First, the segmentation mask is produced using the extracted features and adjusted to the scale of the GT by interpolation. Finally, SSGM is generated by counting the pixels with different values between the segmentation mask and the GT. The wrong segmentation pixels are labeled with 1, and the correct segmentation pixels are labeled with 0. The implementation is as follows:

$$\mathrm{SSGM}(i,j) = \begin{cases} 1, & \mathrm{SM}(i,j) \neq \mathrm{GT}(i,j) \\ 0, & \mathrm{SM}(i,j) = \mathrm{GT}(i,j) \end{cases} \tag{12}$$

where $(i, j)$ represents the corresponding pixel position in both masks. SM is the abbreviation of the segmentation mask. GT is the abbreviation of the GT. Therefore, pixel-wise segmentation cases are obtained to guide the adjustment of restored features in SR precisely.

*b) Semantic weighted mapping:* The imprecise restoration in SR leads to more pixels with wrong segmentation results. However, the spatial information of wrong segmentation pixels is not sensed by SR. In addition, the SR subtask focuses on low-level detail information only. The semantic information is also necessary for the subsequent segmentation. Therefore, a SWM module is proposed to enhance the optimization of SR by utilizing pixel-wise spatial and feature information as guidance. In particular, as shown in Fig. 5, a pixel-wise adjustment of the restored features is performed in SR by utilizing the extracted features of CIs and SSGM from SS. With the help of SWM, the spatial and feature information of specific pixel positions from SS is well integrated to provide more precise guidance which dynamically updates over training time. More details are described as follows.

The SSGM and the features $F_{\mathrm{CI}}$ of CIs from SS are inputted into SWM through feedback connections. The features $F_R$ before the upsampling layer in SR are selected for the mapping. After adjusting SSGM and selected features to the same scale, the features of wrong or correct segmentation pixels $F_*^W$ and $F_*^C$ are distinguished, and the calculation is as follows:

$$\begin{aligned} F_R^W &= \mathrm{SSGM} \cdot F_R \\ F_R^C &= F_R - \mathrm{SSGM} \cdot F_R \\ F_{\mathrm{CI}}^W &= \mathrm{SSGM} \cdot F_{\mathrm{CI}}. \end{aligned} \tag{13}$$

Built on the above analysis, the discrepancy between the restored feature $F_R^W$ and the semantic feature $F_{\mathrm{CI}}^W$ is calculated and utilized as a weighting matrix $W$. The details are as follows:

$$W = 1 - \mathrm{Tanh}\big(\mathrm{abs}\big(F_{\mathrm{CI}}^W - F_R^W\big)\big). \tag{14}$$

Therefore, $F_R^W$ is mapped into the semantic space through the $W$. The enhancement operation is as follows:

$$F_R^{W\prime} = W \cdot F_R^W. \tag{15}$$

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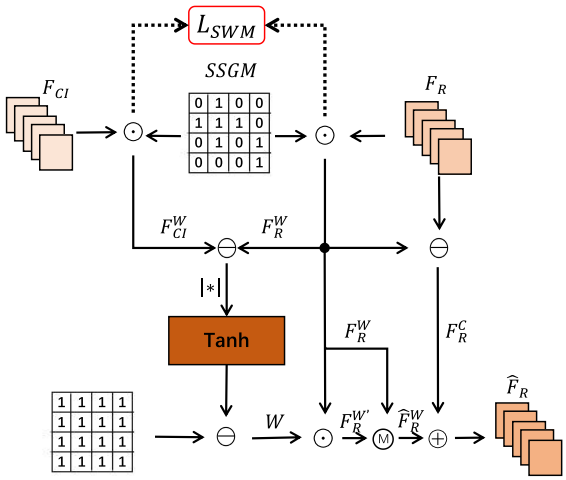



Fig. 5. Details of SWM. The "$M$" and "$-$" symbols in a circle represent the mean value operation and element-wise subtraction, respectively. $|*|$ means taking the absolute value. $F_{\mathrm{CI}}^W$ and $F_R^W$ mean the features of SS extracted from CIs and restored features of SR in wrong segmentation pixel positions, respectively. $F_R^C$ denotes the restored features of SR in correct segmentation pixel positions.

The features of pixes in which the gap between $F_{\mathrm{CI}}^W$ and $F_R^W$ is smaller are given higher weight. Otherwise, less weight is given. In this way, semantic awareness is carried out in the restored features by weighted mapping. Similar to the residual structure design [53], the input feature information is reserved through a short-cut connection. The operation calculation is demonstrated as follows:

$$\hat{F}_R^W = \left(F_R^{W'} + F_R^W\right)/2. \tag{16}$$

The original features $F_R^C$ of correct segmentation pixels are utilized to make up for the final features for the prediction with $\hat{F}_R^W$. The calculation is shown as follows:

$$\hat{F}_R = \hat{F}_R^W + F_R^C. \tag{17}$$

Therefore, the features of pixels with a higher gap are inhibited, which means that the detailed features more related to semantic features are enhanced. The semantic-aware features $\hat{F}_R$ are utilized to generate RIs that contain more indispensable information for the subsequent segmentation. For a right convergence direction, we utilize the similarity loss denoted as $L_{\mathrm{SWM}}$ to make $F_R^W$ more similar to $F_{\mathrm{CI}}^W$. The implementation is as follows:

$$S' = B \times c \times w \times h$$
$$\mathcal{L}_{\mathrm{SWM}} = \frac{1}{S'} \sum_{n}^{B} \sum_{k}^{c} \sum_{i}^{w} \sum_{j}^{h} \left(F_R^W(n,k,i,j) - F_{\mathrm{CI}}^W(n,k,i,j)\right)^2 \tag{18}$$

where $c$, $w$, and $h$ represent the channel number, width, and height of the feature maps, respectively. $S'$ means the total number of pixel positions where the similarity loss is calculated. After this, detailed features are closely related to semantic features in more pixels to achieve better semantic awareness. Therefore, the segmentation is performed in semantic-enhancement RIs. The extracted features are more precise, and excellent segmentation performance is obtained.

*c) PAO for SR:* The original optimization of SR also pays attention to all pixels equally, which is deficient for more adaptive restoration. In the above analysis, wrong segmentation results are caused by the restoration situation, and the optimization of SR in wrong segmentation pixels should be given more attention. Therefore, based on the SSGM containing the spatial information of wrong segmentation pixels, the extra optimization specifically for these pixels is performed. In particular, the original loss function is first reviewed as follows:

$$S'' = B \times C_I \times W \times H$$
$$\mathcal{L}_{\mathrm{MSE}} = \frac{1}{S''} \sum_{n}^{B} \sum_{k}^{C_I} \sum_{i}^{W} \sum_{j}^{H} (\mathrm{RI}(n,k,i,j) - \mathrm{CI}(n,k,i,j))^2. \tag{19}$$

$S''$ means the total number of pixel positions where the restoration loss is calculated. $C_I$ denotes the image channel.

Based on SSGM, the part RI$'$ that needs to be enhanced is separated out, also with the corresponding label CI$'$ as follows:

$$\mathrm{RI}' = \mathrm{RI} \cdot \mathrm{SRGM}$$
$$\mathrm{CI}' = \mathrm{CI} \cdot \mathrm{SRGM}. \tag{20}$$

The loss function for PAO is as follows:

$$\mathcal{L}_{\mathrm{PAO}} = \frac{1}{S''} \sum_{n}^{B} \sum_{k}^{C_I} \sum_{i}^{W} \sum_{j}^{H} (\mathrm{RI}'(n,k,i,j) - \mathrm{CI}'(n,k,i,j))^2. \tag{21}$$

Therefore, there comes the final loss function for SR is as follows:

$$\mathcal{L}_{\mathrm{SR}} = \alpha * \mathcal{L}_{\mathrm{MSE}} + \beta * \mathcal{L}_{\mathrm{PAO}} + \gamma * \mathcal{L}_{\mathrm{SWM}} \tag{22}$$

where $\alpha$, $\beta$, and $\gamma$ represent the weight of the original SR loss, the proposed PAO loss, and the proposed SWM loss in the overall SR loss, respectively. After utilizing the proposed optimization loss, wrong segmentation pixels are also assigned higher weights in the optimization process of SR and given more attention to obtaining more precise optimization for better segmentation.

### C. Iterative Optimization Mode

In the following experiments, the optimization method is analyzed and selected. We propose to optimize the two subtasks iteratively with their loss function when a batch of data comes. When one subtask is optimized, the parameters of another subtask are frozen. There are two subtasks that make up the whole task. It is a bit like multitask learning but not precisely. The original multitask learning usually consists of high-level visual tasks, such as SS, surface normal, and depth estimation tasks. The optimization based on the multitask loss obtains better results for each task with an appropriate weight design. However, SR is an auxiliary task and is utilized to restore images. The optimal solution for SS is pursued finally. The optimization of SS should not be affected by the optimization of SR. As shown in (23), the appropriate weight design is necessary to be achieved

$$\mathcal{L}_{\mathrm{all}} = \lambda * \mathcal{L}_{\mathrm{SS}} + (1-\lambda) * \mathcal{L}_{\mathrm{SR}} \tag{23}$$

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where $\lambda$ represents the coefficient of balance for the two tasks in the multitask loss.

In addition, SR is a low-level visual task. The conflict over the nature of the two subtasks is unknowable which may affect the final segmentation results. Therefore, to avoid potential problems, the optimization method should be studied to find a better manner. Therefore, iteratively optimizing (ITO) the two subtasks is adopted to implement the subsequent experiments.

## IV. Experimental Results and Analyses

In this section, the proposed PW-MFL method is verified by conducting extensive experiments. The detailed implementation information and the experiment results are described in the following.

### A. Datasets

The subsequent experiments are conducted on three widely-used SS datasets, ISPRS Vaihingen, ISPRS Potsdam [54], and iSAID. The details of the three datasets are illustrated in the following.

*1) Vaihingen Dataset:* The Vaihingen dataset includes 33 images of different sizes. Every image is a true orthophoto (TOP). It covers six categories: Impervious surfaces, cars, low vegetation, buildings, trees, and others. The others contain other objects that differ from the first five defined categories. Through cropping the patches from 16 images, we obtain 16 000 subimages with size 512 × 512 for the training of the model, and else 17 images are utilized to verify the model performance.

*2) Postdam Dataset:* There are 38 same-size images in the Potsdam dataset. They are all 6000 × 6000 in size. It has a high resolution of 0.05 m and contains the same class as the Vaihingen dataset. In subsequent experiments, we randomly select 24 images to crop 12 000 subimages with size 512 × 512 to perform the training process, and the remaining 14 images are utilized to verify the training effect.

*3) iSAID Dataset:* The iSAID dataset is the first large instance segmentation dataset in the remote sensing field. It contains 2806 images, including 655 451 object instances in 15 classes. The maximum number of object instances in a single image is 8000 and the average is 239. In this article, we randomly crop images to the size 512 × 512 and only utilize the segmentation mask GT for supervised optimization.

### B. Evaluation Metrics

In subsequent experiments, we utilize the common metrics, mIoU [55], mF1 [56], and OA [57] to demonstrate the performance of the model.

### C. Implementation Details

In the subsequent experiments, the backbone of the segmentation model is chosen as ResNet-101 [53], and the final fully-connected layer is removed.

For ISPRS and iSAID datasets, the total iteration step is set as 40 000 and 80 000 in the training stage, respectively. And the mini-batch of them is set as 4. The basic learning rate is 0.007 and 0.01 for both tasks when training with the Postdam and Vaihingen datasets, respectively. The basic learning rate of the SS task for the iSAID dataset is set as 0.005, and that of the SR task is set as 0.01. The poly decline strategy ($\mathrm{lr} = \mathrm{lr}_{\mathrm{base}} * (1 - (\mathrm{iter}/\max_{\mathrm{iter}}))_{\mathrm{power}}$) is adapted with its power parameter as 0.9. The networks in subsequent experiments are all optimized by a common optimizer, stochastic gradient descent (SGD) with setting the parameter as follows, weight decay $= 1e-4$, momentum $= 0.9$.

For the data, the CIs are with a 512 × 512 size. The capture of the RDIs is through downsampling the CIs as in the previous studies. All subsequent experiments are conducted on one NVIDIA 3090 GPU. Furthermore, the data augmentation operations, random crop, horizontal flip, and scale with a factor from 0.5 to 2.0, are implemented to improve the diversity of the input.

TABLE I
Segmentation Results in RDIs on Vaihingen Dataset

| Method | Degradation Factor | mIoU↑ |
|---|---|---|
| Deeplabv3+ | 1x | **82.34** |
| | 2x | 69.57 |
| | 4x | 48.63 |
| | 8x | 37.01 |

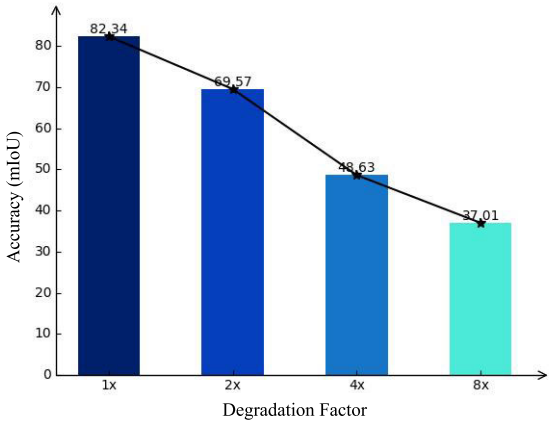



Fig. 6. Accuracy decrease when adopting different degradation factors: 1×, 2×, 4×, and 8×.

### D. Segmentation in RDIs

To intuitively demonstrate the effect of resolution degradation on segmentation results, the experiments are performed when setting different degradation factors, 1× (Origin), 2×, 4×, and 8×. Then, the RDIs are directly used for the segmentation, and the output features are upsampled to obtain segmentation masks with the expected resolution. The values of the evaluation metrics are demonstrated in Table I. Then, the downtrend curve is also shown in Fig. 6 to show the effect brought by the resolution degradation more clearly.

It is evident that as the decline of the resolution is from 1× to 8×, the mIoU drops drastically from 82.34 to 37.01. In addition, when the resolution declines from 2× to 4×, the accuracy drops the most. It is considered that when the decline

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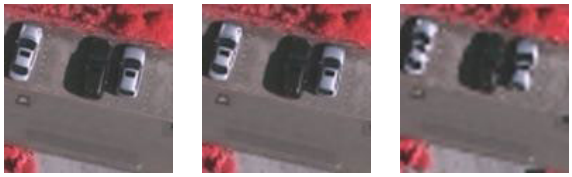



Fig. 7. CIs after degradation with different factors. (a) 1×. (b) 2×. (c) 4×.




Fig. 8. Visualization of SSGM. (a) In the RI. (b) In the CI. (c) Change between (a) and (b). The white positions are wrong segmentation pixel positions and the black positions are correct segmentation pixel positions.

of the resolution produces more impact on the images, the segmentation result gets more restricted. As shown in Fig. 7, from 1× to 2×, the cars in the images are still sharp for relatively accurate segmentation. However, from 2× to 4×, the cars are blurred which is not suitable for the accuracy segmentation.

### *E. Position Correlation Analysis*

We analyze the theoretical feasibility of the novel method in this section. The pixel position correlation between the two tasks is captured to show the relationship between restored results and segmentation results. The relationship between wrong segmentation pixels and upper middle PVD pixels is expected to be obtained to evaluate the effectiveness of the proposed method theoretically. Specifically, the proportion of wrong segmentation pixels belonging to upper middle PVD or lower middle PVD pixels is calculated for the analysis as follows:

$$R = \frac{M_O}{M}. \tag{24}$$

The overall number of pixels in RIs is denoted as $M$. $M_O$ means the counted number of wrong segmentation pixels belonging to upper middle PVD or lower middle PVD pixels. $R$ denotes the proportion. The implementation is conducted based on SSGM and SRGM.

We visualize the SSGM when inputting RIs and CIS as shown in Fig. 8. It can be found that the variation of wrong segmentation pixels between RIs and CIS exists after SR, which represents the complete restoration is not achieved, and the restoration results have an effect on segmentation.

The different PVD cases are classified into two cases, upper middle PVD and lower middle PVD. Then, the proportion is counted and demonstrated in Table II. From the first two rows of the top part of Table II, the proportion of all wrong segmentation pixels belonging to upper middle PVD pixels exceeds that of all wrong segmentation results in lower middle PVD pixels. It shows wrong segmentation pixels are more likely to occur in upper middle PVD pixels; from the last two rows of the top part of Table II, though the proportion of changed wrong segmentation pixels belonging to upper middle PVD pixels declines, it is still higher than that belonging to lower middle PVD pixels.

TABLE II
RESULTS OF THE CALCULATED PROPORTION

| Type | Threshold | R |
|---|---|---|
| **WS in UM** | Mean Value | 11.2 |
| **WS in LM** | Mean Value | 1.6 |
| **VAR WS in UM** | Mean Value | 2.5 |
| **VAR WS in LM** | Mean Value | 0.9 |
| **WS in UM** | Higher than Mean Value | 8.0 |
| **WS in LM** | Higher than Mean Value | 4.0 |
| **VAR WS in UM** | Higher than Mean Value | 2.0 |
| **VAR WS in LM** | Higher than Mean Value | 1.4 |

* WS, VAR WS, UM and LM denote wrong segmentation pixels, variation of wrong segmentation pixels, upper-middle PVD pixels and lower-middle PVD pixels, respectively.

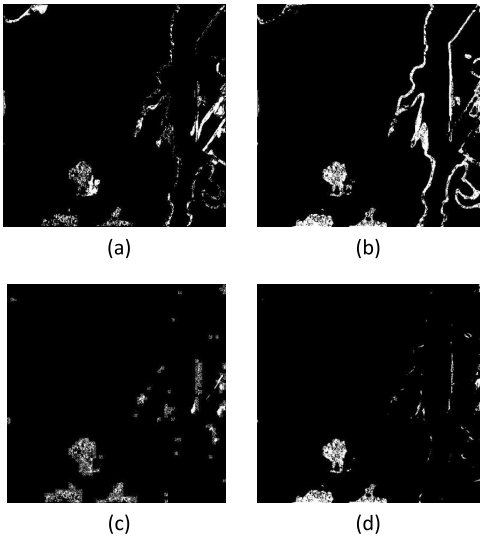



Fig. 9. Visualization of wrong segmentation pixel regions meanwhile belonging to different PVD pixel regions. (a) Belonging to lower middle PVD pixel regions. (b) Belonging to upper middle PVD pixel regions. (c) Changed ones belonging to lower middle PVD pixel regions. (d) Changed ones belonging to upper middle PVD pixel regions.

Moreover, it is found that selecting the threshold value that is higher than the mean value leads to more wrong segmentation positions in lower middle PVD pixels. As the number of wrong segmentation pixels increases, the features of lower middle PVD pixels may also not be effective enough and lose the guidance value. When choosing the threshold value lower than the mean value, the number of lower middle PVD pixels declines, and the guidance is insufficient. Therefore, we utilize the mean value as the threshold value in the subsequent experiments.

Then, we visualize the wrong segmentation pixel regions of the RIs belonging to upper middle PVD or lower middle PVD pixel regions in Fig. 9. It can be seen that there are more wrong segmentation pixels belonging to upper middle PVD pixels.

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TABLE III
EFFECT OF EACH PROPOSED MODULE ON VAIHINGEN DATASET

| Method | PAO | SACC | SWM | IoU per class↑ | | | | | mIoU↑ | OA↑ | mean F1↑ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Imp.surf. | Building | Low veg. | Tree | Car | | | |
| **Degradation Factor (2x)** | | | | | | | | | | | |
| **SRResNet+Deeplabv3+** | × | × | × | 83.93 | 90.31 | 70.02 | 78.53 | 67.11 | 77.98 | 89.55 | 87.37 |
| | ✓ | × | × | **84.35** | 90.78 | 70.63 | 79.19 | 67.89 | 78.55 | **89.57** | 87.73 |
| | × | ✓ | × | 82.28 | **91.50** | 70.35 | 79.75 | 69.91 | 78.76 | 89.22 | 87.89 |
| | × | × | ✓ | 84.23 | 90.54 | 69.66 | 79.07 | 72.36 | 79.17 | 89.71 | 88.17 |
| | ✓ | ✓ | ✓ | 82.68 | 91.41 | **70.82** | **80.14** | **74.70** | **79.95** | 89.52 | **88.69** |
| **Degradation Factor (4x)** | | | | | | | | | | | |
| **SRResNet+Deeplabv3+** | × | × | × | 78.72 | 88.11 | 67.18 | 77.76 | 56.01 | 73.56 | 87.24 | 84.29 |
| | ✓ | × | × | 80.68 | 89.83 | 68.27 | 78.48 | 59.47 | 75.34 | 88.16 | 85.52 |
| | × | ✓ | × | 80.59 | 89.81 | 69.28 | 79.08 | 61.37 | 76.03 | 88.36 | 86.02 |
| | × | × | ✓ | 80.05 | 89.46 | 67.82 | 77.85 | 58.85 | 74.81 | 87.81 | 85.16 |
| | ✓ | ✓ | ✓ | **81.48** | **90.50** | **69.72** | **79.21** | **63.59** | **76.90** | **88.82** | **86.62** |
| **Degradation Factor (8x)** | | | | | | | | | | | |
| **SRResNet+Deeplabv3+** | × | × | × | 71.10 | 80.97 | 61.95 | 74.43 | 29.71 | 63.63 | 83.31 | 76.05 |
| | ✓ | × | × | 78.55 | 85.47 | 64.39 | 74.68 | 38.47 | 68.31 | 86.26 | 79.91 |
| | × | ✓ | × | 79.05 | 86.02 | 64.95 | 75.19 | 41.14 | 69.27 | 86.61 | 80.74 |
| | × | × | ✓ | **79.78** | 86.39 | 65.52 | 75.38 | 41.44 | 69.70 | **86.91** | 81.04 |
| | ✓ | ✓ | ✓ | 78.41 | **87.39** | **66.44** | **76.64** | **47.03** | **71.18** | 86.84 | **82.35** |

According to the above phenomenon, we consider that the upper middle PVD pixels do not provide enough accurate optical information about objects, which poses a huge challenge to uniform feature representation learning for a satisfactory segmentation performance. Therefore, we propose that the features of lower middle PVD pixels can be utilized to enhance the features of upper middle PVD pixels through a SACC module.

Meanwhile, the nonzero proportion of wrong segmentation pixels belonging to lower middle PVD pixels also shows that the PVD situation may not be enough to evaluate the restoration degree for the segmentation-driven SR. More uniform semantic features should be restored in SR. Therefore, we propose that the spatial and semantic information of wrong segmentation positions can be fed to guide SR for more precise restoration.

### *F. Ablation Studies*

In this section, many ablation experiments on the Vaihingen dataset are performed to evaluate each module in the proposed method. Moreover, an in-depth reasonable analysis of each module is also conducted. Deeplabv3+ [19] and SRResNet [38] are utilized as the baseline model. The implementation details are described in the following.

*1) Effect of PW-MFL:* In this section, the effectiveness of each module in the proposed method is proved through experiments, including PAO, SACC, and SWM. For uniform training under different degradation cases, we fixedly set $\alpha = 1$, $\beta = 1$, and $\gamma = 0.3$ in the overall loss function.

As provided in Table III, the degradation parameter is set as 2×, 4×, and 8× to analyze the improvement brought by our method in different resolution cases. As the resolution keeps declining from 2× to 8×, a better improvement of the segmentation results is obtained. It is obvious that the proposed components can bring from 1.97%, 3.34%, and 7.55% improvement under 2×, 4×, and 8× degradation, respectively. It can also be found that more excellent results are obtained especially for the small object, the car class. It is considered that the feature extraction of small objects is more easily affected by the resolution degradation, leading to worse segmentation results. The proposed method can focus on the wrong segmentation pixels so that the car class can get a better improvement. In addition, the segmentation visualization is also demonstrated in Fig. 10, and the segmentation mask is more prominent with using the proposed module.

It is noted that PW-MFL is just implemented in the training stage, and no changes are made for models in the inference stage. Without any increase in computation and memory consumption in the inference stage, PW-MFL can efficiently utilize the whole framework to break down the information barrier between two tasks and exploit the valuable information to achieve better optimization. Compared to the baseline (SR + SS), the final segmentation performance is further boosted without additional computational burden.

*2) Effect of SWM:* The effectiveness of SWM is studied in this section. The feature selection of the middle layers is a random case. The selected features of the shallow or deep layers belong to different levels and may make a diverse impact on the performance of SWM. In addition, considering CIs are also inputted into SS, the deeper features are selected, the more training computation is added. It is expected to accelerate the training stage and obtain better results. Therefore, the experiments of selecting features in different layers are performed under one degradation case, 4×. The corresponding experimental results are provided in Table IV.

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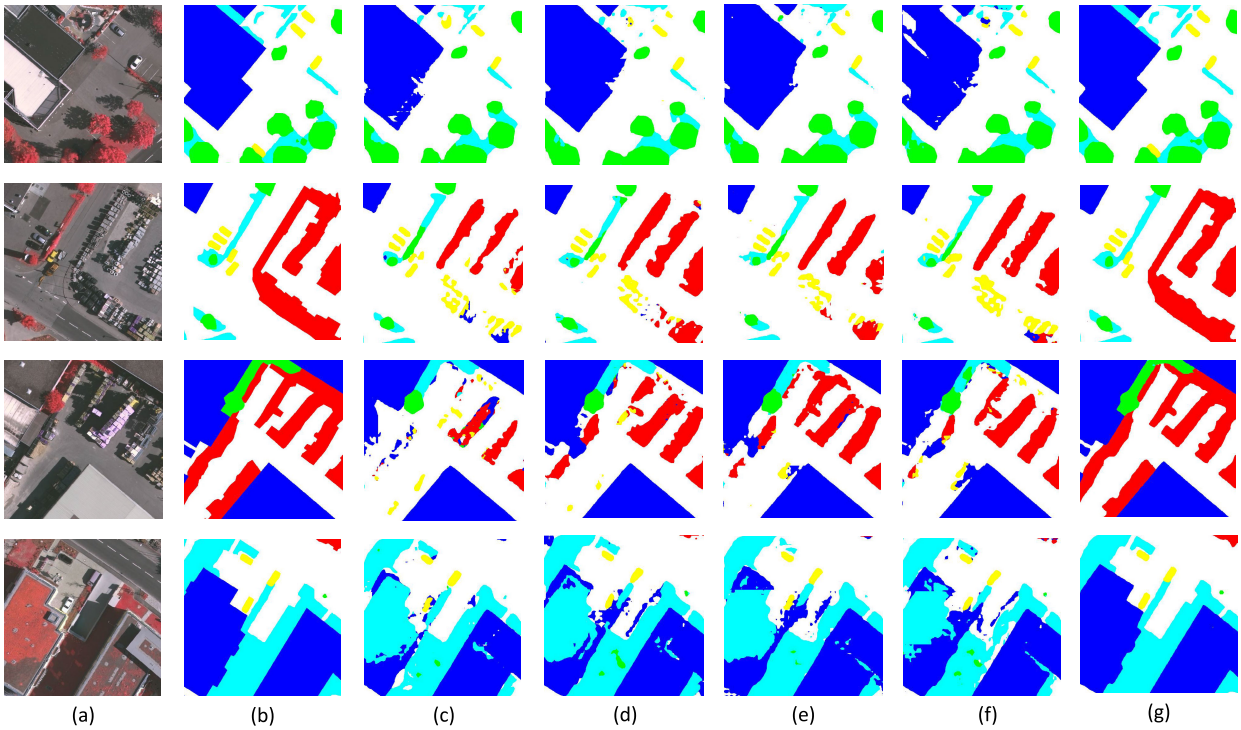



Fig. 10. 4× Segmentation masks visualization on Vaihingen datasets. (a) Input images. (b)–(g) In turn, GT, without proposal work, with PAO, with SACC, with SWM, and with all proposed modules.

TABLE IV
RESULTS OF THE FEATURE SELECTION ON THE VAIHINGEN DATASET

| Layer | FlOPs(G) ↓ | mIoU ↑ | PSNR ↑ | SSIM ↑ |
|---|---|---|---|---|
| **Degradation Factor (4x)** | | | | |
| None | – | 73.56 | 25.32 | 0.75 |
| Conv1 | 0.057 | 73.69 | 26.06 | 0.76 |
| Conv2 | 1.208 | 74.16 | 26.37 | 0.77 |
| Conv3 | 2.416 | **74.81** | **26.45** | **0.77** |
| layer1 | 1.976 | 73.90 | 26.26 | 0.76 |
| layer2 | 2.789 | 74.48 | 26.34 | 0.77 |

TABLE V
EXPLORATION OF THE EFFECTIVENESS OF USING SSGM IN SWM

| method | mIoU↑ | OA↑ | mean F1↑ |
|---|---|---|---|
| SRResNet+Deeplabv3+ (Basic) | 73.56 | 87.24 | 84.29 |
| +SWM (without SSGM) | 74.68 | 87.50 | 85.13 |
| +SWM (with SSGM) | **74.81** | **87.81** | **85.16** |

From Table IV, considering there also comes a compound loss for optimizing SR based on the new loss designed in SWM, the weight distribution may have a negative impact on the results. After utilizing the **Conv1** even behaves worse than the baseline. As the depth increases, the mIoU becomes better and the computational complexity also increases. Therefore, the depth can be chosen and tested according to the experiments for the balance between speed and accuracy. In this article, the feature maps after **Conv3** are selected for SWM.

In addition, the original feature attention in SR is given more to low-level details as Fig. 11(b). The feature restoration may ignore the semantic features, which may be unsuitable for the subsequent segmentation. After using SWM, the feature attention in SR is more inclined to high-level semantic information like in SS as shown in Fig. 11(c) and (d). A better feature restoration is conducted, and the segmentation results in RIs are improved. Moreover, from the SR aspect, the restoration quality of RIs is also evaluated via two common indicators: PSNR and SSIM to show the effect of using SWM. The obtained values are also recorded in Table IV. The PSNR and SSIM also increase after using the designed module. The restoration performance is also improved. Therefore, the high-level semantic information is incorporated into the restoration process as guidance to obtain better results.

Moreover, we analyze the performance of using SSGM in SWM. The corresponding results are provided in Table V. It can be found that after using SSGM in SWM, the evaluation indicators of mIoU, OA, and mean F1 can be further improved by 0.13%, 0.31%, and 0.03%, respectively. The results show the necessity and effectiveness of the usage of SSGM in SWM. The more precise semantic restoration is achieved and suitable for segmentation by using SSGM in SWM.

There also comes an analysis of feature uniformity between the two tasks in wrong segmentation pixel regions. A sample of 12 randomly selected images is utilized for the analysis. Then, before and after using SWM, the feature uniformity in wrong segmentation pixel regions between SR and SS is evaluated by analyzing the Euclidean distance as the loss $L_{\text{SWM}}$.

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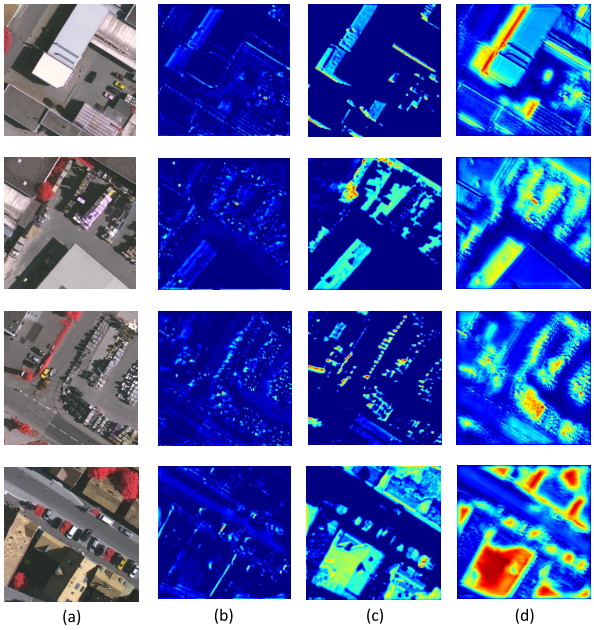



Fig. 11. Comparison between feature attention visualization in SR and SS. (a) Input image. (b) Attention in SR without using SWM. (c) Attention in SS. (d) Attention in SR with using SWM.

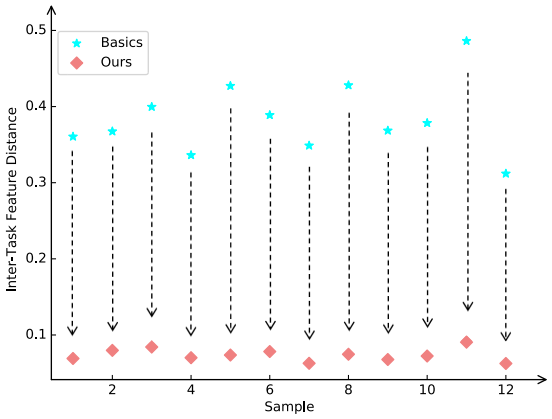



Fig. 12. Visualization of intertask feature distance in selected samples.

There is the visualization of the analysis result in Fig. 12. The intertask feature distance is reduced after using SWM. The distance is narrowed from around 0.4 to within 0.1. The intertask feature uniformity is improved after using the SWM. The more semantic information is incorporated into SR, and better segmentation results are achieved.

*3) Effect of SACC:* In this section, the analysis of the SACC module is carried out under one degradation case, 4×. There come qualitative experiments to demonstrate the effectiveness of the module. According to the extracted features, we analyze the attention variation of SS in the upper middle PVD pixels. The visualization of the results is shown in Fig. 13.

It can be found that the relationship between upper middle PVD pixels and lower middle PVD pixels is improved in pixels with the same class after using SACC. The upper middle PVD pixels obtain more attention, and the attention in pixels within the same class is distributed more uniformly. The features of upper middle PVD pixels are enhanced.

TABLE VI
EXPLORATION OF THE WEIGHT OF PAO

| $\beta$ | mIoU↑ | OA↑ | mean F1↑ |
|---|---|---|---|
| baseline | 73.56 | 87.24 | 84.29 |
| 0.5 | 76.62 | 88.66 | 86.44 |
| 1 | 75.34 | 88.16 | 85.52 |
| 1.5 | 76.40 | 88.60 | 86.28 |
| 2 | **77.13** | **88.79** | **86.79** |
| 2.5 | 76.01 | 88.42 | 86.02 |
| 3 | 76.63 | 88.56 | 86.45 |
| 3.5 | 76.79 | 88.65 | 86.56 |

TABLE VII
PERFORMANCE OF DIFFERENT OPTIMIZATION MODES ON VAIHINGEN DATASET

| Method | JOO | INO | ITO | mIoU ↑ |
|---|---|---|---|---|
| **Degradation Factor (4x)** | | | | |
| **SRResNet+Deeplabv3+(Basic)** | ✓ | × | × | **74.08** |
| | × | ✓ | × | 72.74 |
| | × | × | ✓ | 73.56 |
| **SRResNet+Deeplabv3+(Ours)** | ✓ | × | × | 76.28 |
| | × | ✓ | × | – |
| | × | × | ✓ | **76.90** |

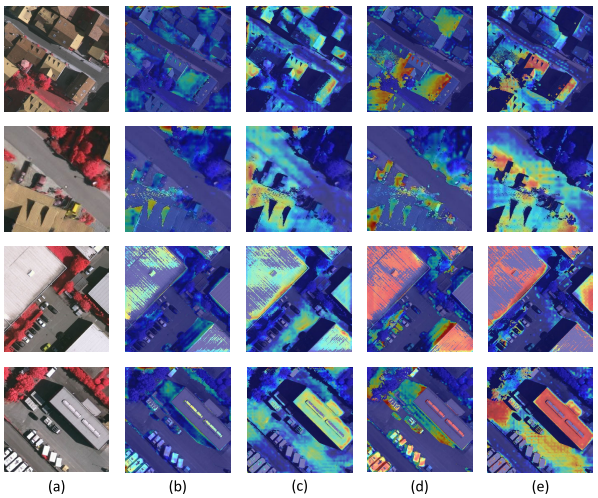



Fig. 13. Visualization of the attention in SS. (a) Input image. (b) Baseline in upper middle PVD pixels. (c) Baseline in lower middle PVD pixels. (d) Proposed method in upper middle PVD pixels. (e) Proposed method in lower middle PVD pixels.

*4) Effect of PAO:* In this section, we carry out experiments on the Vaihingen dataset to analyze the effect of different weights $\beta$ of PAO for final segmentation results. SRResNet and Deeplabv3+ are chosen as the baseline for the relevant experiments, where only the PAO module is considered with various weight $\beta$ settings and fixed degradation factor 4×. The corresponding results are provided in Table VI.

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TABLE VIII
PERFORMANCE ON VAIHINGEN DATASET

| Method | IoU per class↑ | | | | | mIoU↑ | OA↑ | mean F1↑ |
|---|---|---|---|---|---|---|---|---|
| | Imp.surf. | Building | Low veg. | Tree | Car | | | |
| **Degradation Factor (4x)** | | | | | | | | |
| **SRResNet+Deeplabv3+ (basic)** | 78.72 | 88.11 | 67.18 | 77.76 | 56.01 | 73.56 | 87.24 | 84.29 |
| **SRResNet+Deeplabv3+ (ours)** | **82.56** | **91.02** | **70.31** | **79.61** | **69.00** | **78.50** | **89.30** | **87.72** |
| **SRResNet+PSPNet (basic)** | 80.81 | 90.62 | 67.60 | 78.07 | 59.50 | 75.32 | 88.26 | 85.48 |
| **SRResNet+PSPNet (ours)** | **82.15** | **91.01** | **69.44** | **79.51** | **66.25** | **77.67** | **89.08** | **87.15** |
| **EDSR+PSPNet (basic)** | 80.70 | 89.70 | 68.04 | 78.13 | 59.61 | 75.23 | 88.06 | 85.46 |
| **EDSR+PSPNet (ours)** | **82.32** | **90.25** | **70.44** | **78.88** | **65.69** | **77.52** | **88.98** | **86.52** |
| **DDBPN+CCNet (basic)** | 79.90 | 88.78 | 68.50 | 78.72 | 59.57 | 75.09 | 87.88 | 85.39 |
| **DDBPN+CCNet (ours)** | **82.18** | **90.14** | **70.26** | **78.78** | **64.55** | **77.18** | **88.90** | **86.83** |
| **RCAN+EANet (basic)** | 81.58 | 90.59 | 69.49 | 78.09 | 65.91 | 77.13 | 88.65 | 86.81 |
| **RCAN+EANet (ours)** | **82.90** | **91.48** | **70.39** | **79.79** | **67.50** | **78.41** | **89.43** | **87.64** |

Compared with that without using PAO ($\beta = 0$), the baseline with PAO achieves better segmentation under different weights. This validates the effectiveness and robustness of the proposed PAO module. Moreover, the optimal setting of the weight ($\beta = 2$) is found under the 4× degradation case. The corresponding optimal segmentation performance is obtained, namely, 77.18% mIoU.

*5) Effect of Iterative Optimization Mode:* In this section, there is a comparison of three optimization modes for training. First, the losses of the SR and the SS are jointly optimized (JOO). Second, individually optimize (INO) each task: initially optimize the SR well and then optimize the SS based on the well-RIs. Third, the utilized optimization mode: the two tasks are iteratively optimized (ITO) in one step of the training stage. When one task is optimized, the parameters of the other task are frozen.

As shown in Table VII, the joint optimization of the baseline is better than the other two methods, which enhances the generalization ability of the whole model. However, the ITO mode can get an optimal result after using the proposed method. When the final loss objective becomes more complex, it is more difficult to conduct an appropriate weight distribution. The unfit weight distribution even produces a bad effect. The optimization of SS is influenced by the optimization of SR. Therefore, the optimization mode should be selected by performing experiments.

### G. Generalization Studies

In this section, the effectiveness and generalization of PW-MFL are validated on three datasets, ISPRS Vaihingen, ISPRS Potsdam, and iSAID, by utilizing well-behaved and widely-used models of SR and SS to form various baselines. The subsequent experiments are conducted when the degradation factor is fixedly set as 4×. Besides the selected model combination, SRResNet [38] + Deeplabv3+ [19] in ablation studies, another four model combinations, SRResNet [38] + PSPNet [7], EDSR [33] + PSPNet [7], DDBPN [35] + CCNet [6], and RCAN [36] + EANet [23] are formed to validate the proposed PW-MFL. The final state-of-the-art (SOTA) segmentation results for each class and the overall segmentation results are obtained and provided in the following content.

*1) Performance on Vaihingen Dataset:* In this section, extensive experiments are conducted on the Vaihingen dataset with five model combinations. The final segmentation results are demonstrated in Table VIII.

From the results in Table VIII, all model combinations get different levels of performance improvements after using the proposed method, and the average improvement is 2.59% mIoU. The model combination of SRResNet + Deeplabv3+ achieves the best segmentation result of 78.50% mIoU, meanwhile getting a maximal improvement of 4.94% mIoU with using the proposed method. The PW-MFL can be easily expanded to various model combinations to achieve more satisfactory segmentation results, which has a satisfactory generalization.

*2) Performance on Potsdam Dataset:* To validate the effectiveness and generalization of the proposed method, we carried out experiments on the widely used Potsdam benchmark. The numerical comparisons before and after using the proposed method with five model combinations are detailed in Table IX.

As indicated in Table IX, all model combinations get different levels of improvement in segmentation results and the improvements are prominent. The model combination of EDSR + PSPNet gets the maximum improvement of 5.17% mIoU while achieving the best segmentation result of 83.46% mIoU. Moreover, the segmentation masks are clearly shown in Fig. 14. It is observed that after using our method, the segmentation masks become more precise. In addition, a better segmentation performance is obtained in the small object, and the results in edge positions are also more excellent. The initial purpose is achieved by implementing the proposed method.

*3) Performance on iSAID Dataset:* We further conduct experiments on another common segmentation dataset, the iSAID dataset to evaluate the generalization of our proposed method. The corresponding experimental results based on five model combinations are provided in Table X.

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TABLE IX
PERFORMANCE ON POTSDAM DATASET

| Method | IoU per class↑ | | | | | mIoU↑ | OA↑ | mean F1↑ |
|---|---|---|---|---|---|---|---|---|
| | Imp.surf. | Building | Low veg. | Tree | Car | | | |
| **Degradation Factor (4x)** | | | | | | | | |
| **SRResNet+Deeplabv3+ (basic)** | 81.80 | 91.32 | 69.63 | 63.23 | 88.91 | 78.97 | 86.18 | 87.83 |
| **SRResNet+Deeplabv3+ (ours)** | **85.42** | **91.81** | **72.53** | **73.93** | **90.01** | **82.74** | **88.73** | **90.34** |
| **SRResNet+PSPNet (basic)** | 80.58 | 89.83 | 66.92 | 70.62 | 88.86 | 79.36 | 85.81 | 88.19 |
| **SRResNet+PSPNet (ours)** | **85.35** | **91.94** | **73.84** | **74.66** | **91.12** | **83.38** | **89.06** | **90.73** |
| **EDSR+PSPNet (basic)** | 82.57 | 87.72 | 68.35 | 66.02 | 87.77 | 78.29 | 85.95 | 87.51 |
| **EDSR+PSPNet (ours)** | **85.99** | **92.68** | **73.43** | **74.68** | **90.53** | **83.46** | **89.26** | **90.78** |
| **DDBPN+CCNet (basic)** | 83.18 | 89.86 | 69.92 | 70.59 | 89.48 | 80.61 | 87.20 | 89.00 |
| **DDBPN+CCNet (ours)** | **84.99** | **91.95** | **72.78** | **73.84** | **90.95** | **82.93** | **88.73** | **90.45** |
| **RCAN+EANet (basic)** | 84.01 | 90.83 | 72.31 | 72.60 | 87.83 | 81.52 | 87.89 | 89.62 |
| **RCAN+EANet (ours)** | **85.03** | **92.12** | **73.06** | **74.24** | **91.32** | **83.15** | **88.73** | **90.58** |

TABLE X
PERFORMANCE ON ISAID DATASET

| Method | IoU per class↑ | | | | | | | | | | | | | | | mIoU ↑ | OA ↑ | mean F1↑ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | PL | BD | BR | GTF | SV | LV | SH | TC | BC | ST | SBF | RA | HA | SP | HC | | | |
| **Degradation Factor (4x)** | | | | | | | | | | | | | | | | | | |
| **SRResNet+Deeplabv3+ (basic)** | 71.17 | 45.24 | **22.14** | 32.19 | 30.07 | 48.81 | **53.22** | 51.07 | **25.49** | 29.96 | 46.02 | 27.07 | **46.38** | **40.29** | 9.61 | 38.58 | **91.51** | 54.00 |
| **SRResNet+Deeplabv3+ (ours)** | **74.22** | **49.61** | 20.92 | **49.38** | **31.95** | **52.51** | 50.60 | **72.87** | 22.56 | **50.02** | **53.75** | **32.55** | 44.96 | 40.17 | **18.86** | **44.33** | **91.51** | **59.64** |
| **SRResNet+PSPNet (basic)** | 67.54 | 61.49 | 16.84 | 36.10 | 29.43 | 37.49 | 51.89 | 43.76 | 38.04 | 38.58 | 36.72 | 5.85 | 48.91 | 39.76 | 0.82 | 36.88 | 90.06 | 51.18 |
| **SRResNet+PSPNet (ours)** | **72.93** | **61.73** | **22.96** | **38.67** | **30.98** | **47.71** | **48.95** | **68.26** | **39.80** | **48.61** | **44.11** | **40.91** | **46.29** | **39.23** | **14.59** | **44.38** | **90.33** | **59.95** |
| **EDSR+PSPNet (basic)** | 72.78 | **62.09** | **23.38** | 34.86 | 29.54 | **48.89** | **53.73** | 60.93 | 34.39 | 42.98 | 37.97 | 44.98 | **45.57** | 6.13 | 8.18 | 40.43 | **90.31** | 55.00 |
| **EDSR+PSPNet (ours)** | **75.05** | 60.79 | 20.74 | **35.34** | **32.57** | 43.40 | 51.81 | **68.62** | **43.98** | **56.41** | **48.74** | **48.52** | 37.15 | **14.80** | **14.67** | **43.51** | 89.35 | **58.46** |
| **DDBPN+CCNet (basic)** | 69.51 | 64.10 | **26.35** | 14.38 | **26.67** | **49.74** | 50.06 | 70.15 | 52.66 | 48.06 | 2.10 | 32.54 | 41.00 | 21.10 | **24.54** | 39.46 | 92.91 | 53.54 |
| **DDBPN+CCNet (ours)** | **69.97** | **66.83** | 26.11 | **23.34** | 22.91 | 46.78 | **50.96** | **73.19** | **53.76** | **49.42** | **29.13** | **39.37** | **42.28** | **25.04** | 18.23 | **42.49** | **93.38** | **57.52** |
| **RCAN+EANet (basic)** | **74.78** | 50.20 | 23.92 | 36.95 | 21.58 | 30.14 | 52.16 | 75.78 | **45.32** | **58.58** | 45.23 | **46.93** | 44.08 | 30.22 | **17.50** | 43.56 | 91.61 | 58.76 |
| **RCAN+EANet (ours)** | 73.20 | **52.60** | **23.93** | **47.14** | **24.93** | **52.38** | **54.86** | **81.80** | 38.58 | 57.13 | **47.33** | 40.17 | **46.95** | **42.54** | 16.78 | **46.69** | **92.62** | **61.86** |

* The 15 classes of iSAID dataset: plane (PL), baseball diamond (BD), bridge (BR), ground track field (GTF), small vehicle (SV), large vehicle (LV), ship (SH), tennis court (TC), baseball court (BC), storage tank (ST), soccer ball field (SBF), roundabout (RA), harbor (HA), swimming pool (SP) and helicopter (HC).

All of the selected model combinations achieve prominent improvements by utilizing the proposed method. DDBPN + CCNet gets the lowest improvement of 3.03% mIoU, and SRResNet + PSPNet gains the highest improvement of 7.50% mIoU. In addition, RCAN + EANet achieves the best performance of 46.69% mIoU by using our proposed method. The visualization variations of segmentation performance are also shown in Fig. 15. As we can see, after using the proposed method, the segmentation mask becomes more satisfactory, especially in object edge regions. Through the proposed modules, the intraclass uniformity between regions with different restoration cases is enhanced, and the semantic-related texture features are restored in RIs. It is more distinguishable between objects and between objects and background regions. The extracted feature representations for segmentation are more satisfactory, and better segmentation performance is achieved.

### *H. Applicability Discussion*

There exist many factors that affect the spatial resolution of real images, such as ground-projected sample interval (GSI), sensor noise, nonuniform targets and backgrounds, and variable solar angle and topography [58]. The proposed learning

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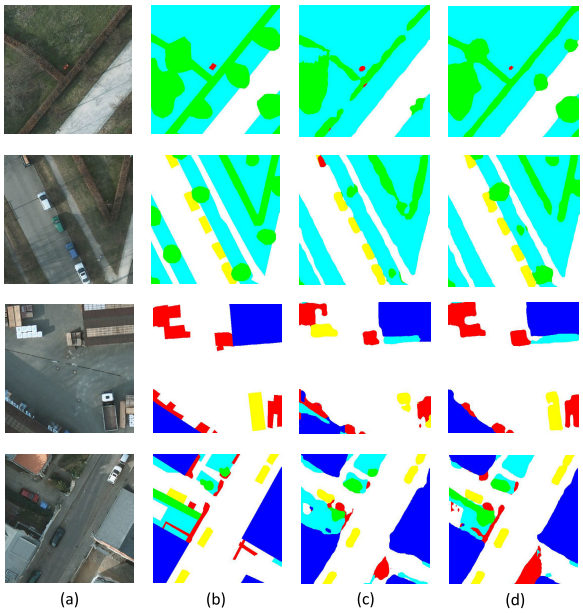



Fig. 14. Visualization of segmentation masks on Potsdam test set when the degradation factor is set as 4×. (a) Input image. (b) GT. (c) and (d) Segmentation masks obtained by the baseline before and after using the proposed method, respectively.

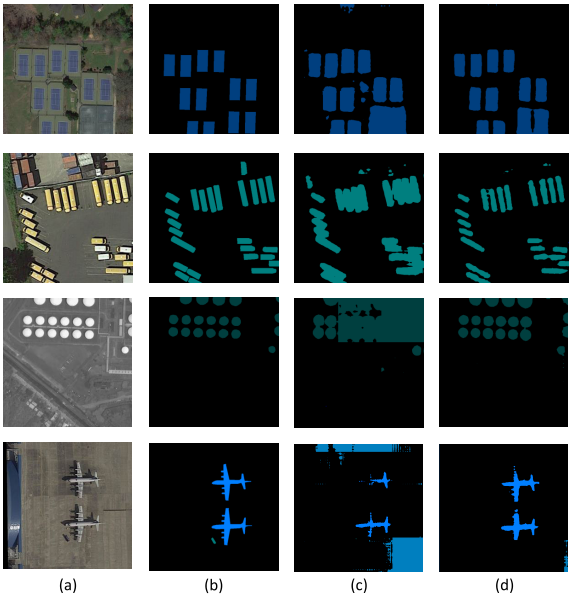



Fig. 15. Visualization of segmentation masks on iSAID value set when the degradation factor is set as 4×. (a) Input image. (b) GT. (c) and (d) Segmentation masks obtained by the baseline before and after using the proposed method, respectively.

method aims to promote the segmentation performance of RDIs that just result from insufficient GSI. Therefore, the scope of application is still relatively limited. In the future, more cases caused by other degradation factors should be considered based on the proposed PW-MFL method, and more effective information guidance should be explored to improve the final segmentation results.

## V. Conclusion

We propose a novel learning method to perform SS in resolution-degraded aerial images with SR aid in this article. It is named as PW-MFL. Built on the original open-loop structure, additional feed-forward and feedback connections are designed between SR and SS to make the two subtasks more tightly coupled. One subtask is optimized based on the guidance from another subtask through these connections. The additional pixel-wise spatial information is transferred to each other to perform semantic-aware restoration and learn better feature representation in specific pixel positions for better segmentation, respectively. The experiments on ISPRS Vaihingen, ISPRS Postdam, and iSAID datasets are carried out. The results prove the effectiveness and generalization of the novel method. The 2%–8% mIoU improvement is achieved as the degradation factor increases. Moreover, better performance is shown especially in object edge regions. In future work, more effective and precise information will be studied, and reducing the model complexity will be considered.

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**Jinze Yang** received the B.Sc. degree from Xidian University, Xi'an, China, in 2020. He is currently pursuing the Ph.D. degree with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China.

His research interests include deep learning, remote sensing image interpretation, and mutual-feed learning.

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**Youming Wu** received the B.Sc. and Ph.D. degrees from Beihang University, Beijing, China, in 2013 and 2020, respectively.

He is an Associate Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing. His research interests include remote sensing image interpretation and the improvement of spaceborne SAR image quality.



**Wei Dai** received the B.Sc. degree from Xidian University, Xi'an, China, in 2016. He is currently pursuing the Ph.D. degree with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China.

His research interests include computer vision, pattern recognition, and remote sensing image processing, especially fine-grained visual classification.



**Wenhui Diao** received the B.Sc. degree from Xidian University, Xi'an, China, in 2011, and the Ph.D. degree from the Institute of Electronics, Chinese Academy of Sciences, Beijing, China, in 2016.

He is currently an Associate Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences. His research interests include computer vision and remote sensing image analysis.



**Zining Zhu** (Member, IEEE) received the B.Sc. degree from Jilin University, Changchun, China, in 2020. She is currently pursuing the Ph.D. degree with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China.

Her research interests include deep learning, remote sensing image interpretation, and few-shot incremental learning.



**Xian Sun** (Senior Member, IEEE) received the B.Sc. degree from the Beijing University of Aeronautics and Astronautics, Beijing, China, in 2004, and the Ph.D. degree from the Institute of Electronics, Chinese Academy of Sciences, Beijing, in 2009.

He is currently a Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences. His research interests include computer vision, geospatial data mining, and remote sensing image understanding.



**Kun Fu** (Member, IEEE) received the B.Sc., M.Sc., and Ph.D. degrees from the National University of Defense Technology, Changsha, China, in 1995, 1999, and 2002, respectively.

He is currently a Professor with the Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing, China. His research interests include computer vision, remote sensing image understanding, and geospatial data mining and visualization.

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